One-Handed and Off-Balance: Does Proximity Feedback Modality Reduce Falls in a Smartphone Tilt Task?

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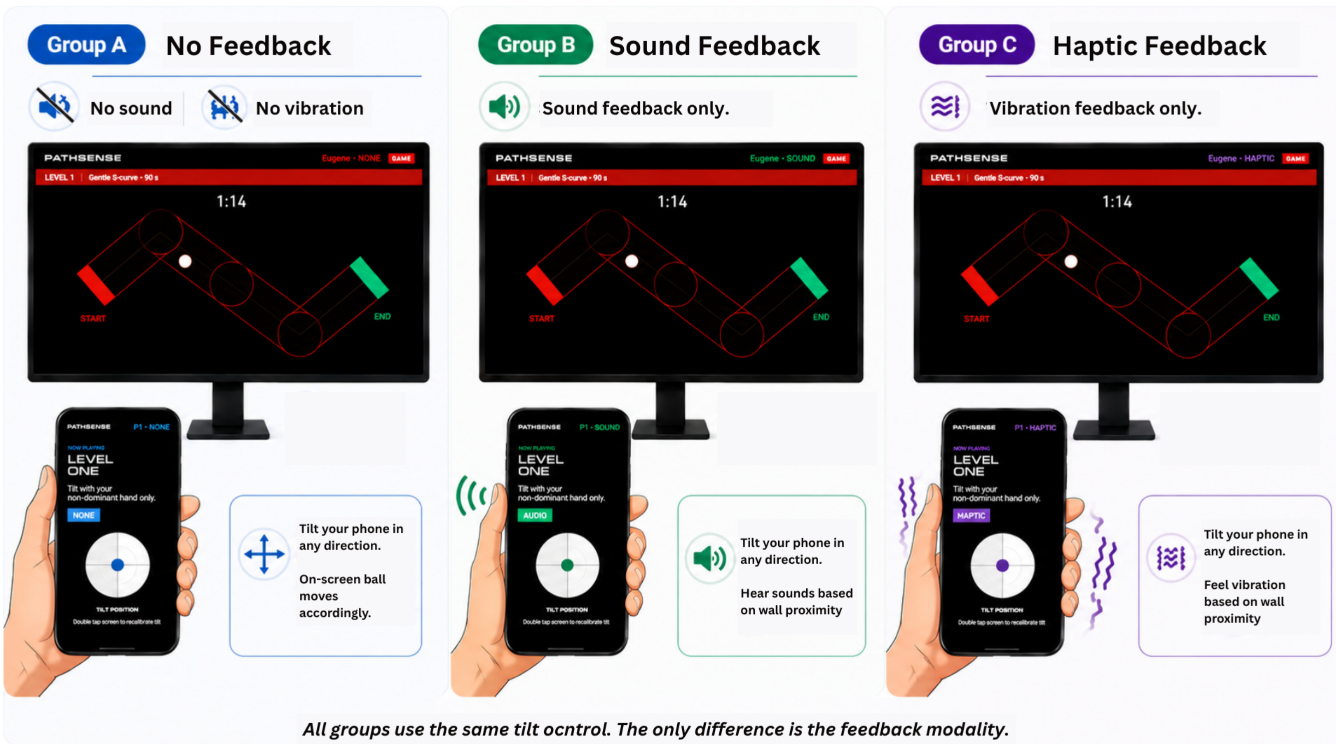

Figure 1: The three between-subjects feedback conditions in PathSense. All groups use the same tilt control; the only difference is the feedback modality.

Abstract

Tilt-based smartphone interfaces are ubiquitous, yet they often feel clumsy when controlled with the non-dominant hand. We hypothesize that this reflects a sensory calibration gap that proximity-based feedback could help close. We present PathSense, a smartphone-native tightrope tilt task comparing three feedback modalities (haptic, audio, none) with 25 participants across four difficulty levels. In this preliminary study, no modality showed a significant advantage in fall reduction (Kruskal-Wallis, all $p > 0.19$); however, change-from-baseline analysis reveals a critical confound in baseline practice performance (audio: 0.33 falls vs. haptic: 4.88, $p < 0.05$) that warrants controlled follow-up. We discuss implications for modality design in motor augmentation and provide effect size estimates for a larger planned follow-up ($n \approx 50$).

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CCS Concepts

• **Human-centered computing → HCI theory, concepts and models**; *Haptic devices.*

Keywords

body schema, haptic feedback, motor learning, non-dominant hand, proximity feedback, smartphone, tilt interaction

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1 Introduction

Tilt-based interfaces require real precision from the non-dominant hand, the limb we typically rely on less, yet one that already leans heavily on peripheral proprioceptive feedback rather than fine cortical resolution [1]. When we actively use a tool, the brain extends its spatial self-model to encompass it [2]; if the held phone is already encoded as a bodily extension, proximity warnings delivered *through* it may update the user's spatial model more directly than warnings on a separate channel. Whether feedback *modality* shapes that benefit is less clear: haptic cues act on the same sensorimotor channel as the task, while audio cues are processed on a separate channel and require deliberate signal-to-correction mapping [3]. Existing evidence comes from lab-grade rehabilitation rigs [4]; whether smartphone-delivered proximity feedback produces comparable effects in a non-dominant hand tilt task remains untested.

We present *PathSense*, a smartphone tightrope tilt task in which participants navigate a ball along a narrow pre-designed path using only their non-dominant hand, receiving graded proximity warnings in one of three modalities (haptic, audio, or none), shown in Figure 1. Falls (ball-off-path events) are the primary outcome. We address four research questions: (RQ1*) does modality affect fall rate? (RQ2) does any feedback reduce falls vs. control? (RQ3) which modality performs better? (RQ4) does improvement generalise across difficulty levels?

2 Related Work

Motor augmentation through external feedback has its roots in cybernetics and human factors research. Sigrist et al. (2013) found that haptic feedback excels at rapid online correction due to its colocation with the acting limb, while auditory feedback may better support learning in tasks requiring complex mapping [3]. In rehabilitation, haptic guidance has shown consistent gains for proprioceptive training [4], yet audiomotor coupling is well-documented in music and sports contexts [**?**]. Regarding body schema, Lin et al. (2022) provided behavioural evidence that smartphone use updates body schema similarly to classic tool integration [5], suggesting that feedback delivered *through* a held phone may have privileged access to this system. Prior work on non-dominant hand control has focused on bimanual coordination [6], but not on proximity warnings in smartphone tilt; PathSense bridges this gap.

3 Study Design and Method

Participants. We recruited 25 participants (mean age 14.9, SD 1.8; 9 female; 19 right-handed) with no prior tilt-control experience required. Participants were assigned in counterbalanced order to haptic ($n = 8$), audio ($n = 9$), or none ($n = 8$) groups; groups did not differ significantly on age ($p = 0.37$) or gaming experience ($p = 0.41$).

Apparatus and Task. Participants held an Android smartphone with their non-dominant hand and tilted it to keep a ball centred on a narrow path displayed on an external monitor. Lateral tilt (gamma) drove horizontal acceleration; forward-back tilt (beta) drove vertical acceleration. A *fall* occurred when the ball left the path corridor; the ball then respawned at the most recent checkpoint.

Conditions and Procedure. The study was between-subjects (Table 1). Feedback escalated in real time as the ball approached the path edge: haptic pulses increased in intensity and frequency; audio tone pitch and rate escalated; the none condition provided no feedback. After each round, participants completed a brief post-round survey.

Table 1: Difficulty levels and task parameters

Level	Path shape	Duration	Feedback
Practice (L1)	S-curve	90 s	None (all groups)
Easy (L2)	S-curve	90 s	Modality-dependent
Medium (L3)	N-shape	75 s	Modality-dependent
Hard (L4)	Tight zigzag	60 s	Modality-dependent

Figure 2: Study setup: participant holds the phone with the non-dominant hand and tilts to keep the ball on-path. The laptop displays the game in real time; the tablet shows the admin panel.

Data Collection and Analysis. Game data were logged to an SQLite database via a Node.js server; trajectory data were recorded at 10 Hz (ball position, proximity level, gamma/beta tilt angles). Rounds with missing `round_duration_ms` were excluded (4 of 112 expected rounds), yielding 100 rounds for analysis (25 sessions × 4 levels minus 8 excluded). Non-parametric Kruskal-Wallis H tests assessed modality effects at each experimental level, followed by pairwise Mann-Whitney U tests with Holm correction; effect sizes are rank-biserial r. Change-from-baseline deltas ($\Delta = \text{falls}_{\text{level}} - \text{falls}_{\text{practice}}$) controlled for heterogeneous baseline performance.

4 Results

Primary analysis: raw falls. Table 2 shows mean falls by modality and level. Kruskal-Wallis tests on experimental levels showed no significant modality effect: Easy $H(2) = 3.294$, $p = 0.193$; Medium $H(2) = 0.600$, $p = 0.741$; Hard $H(2) = 0.477$, $p = 0.788$. Pairwise Mann-Whitney tests (Holm-corrected) confirmed no significant differences at any level (all $p > 0.27$; rank-biserial r from -0.44 to $+0.25$). To quantify evidence for the null, Bayes factors strongly favour no modality difference: $BF_{01} \approx 83.8$ at Easy, 7.0 at Medium, 5.4 at Hard. A winsorised sensitivity analysis (95th percentile per cell) reproduced the same pattern (all $p > 0.16$).

Table 2: Mean (SD) falls by modality and difficulty level. Practice is the pre-feedback baseline (no feedback given to any group). Group sizes: Haptic $n = 8$, Audio $n = 9$, None $n = 8$.

Level	Haptic ($n = 8$)	Audio ($n = 9$)	None ($n = 8$)
Practice	4.88 (7.34)	0.33 (0.50)	3.00 (4.17)
Easy	3.50 (5.71)	0.33 (0.71)	0.62 (0.74)
Medium	3.62 (6.39)	2.44 (3.81)	3.62 (6.32)
Hard	13.00 (4.04)	11.78 (7.36)	13.25 (5.18)

Baseline confound. A critical asymmetry emerged in the practice round (L1), where no feedback was given: audio participants averaged 0.33 falls (SD 0.50) while haptic averaged 4.88 (SD 7.34) and none averaged 3.00 (SD 4.17), representing a 10-fold gap (Mann-Whitney $U = 28$, $p = 0.04$, $r = -0.57$). This suggests pre-existing group differences in baseline tilt control. Change-from-baseline deltas show that at Easy level haptic improved by -1.38 falls, none by -2.38, and audio remained flat (0.00); at Hard level all groups deteriorated by +8 to +11 falls with no modality advantage.

Steering response. To investigate real-time responses to proximity warnings, we identified 2,372 proximity escalation events (consecutive frames where `proximity_level` increased) and extracted a ±4 s window around each, excluding events within 500 ms of a fall or respawn. Figure 3 shows time-locked mean proximity level and tilt magnitude per modality. Median time-to-recovery was similar across groups: haptic 400 ms, audio 416 ms, none 417 ms; 98–99% of escalations recovered within 3 s. Tilt reversal frequency within 500 ms differed slightly (audio 43.0%, none 32.6%, haptic 27.2%), yet all modalities showed comparable proximity return to baseline within 1–2 s.

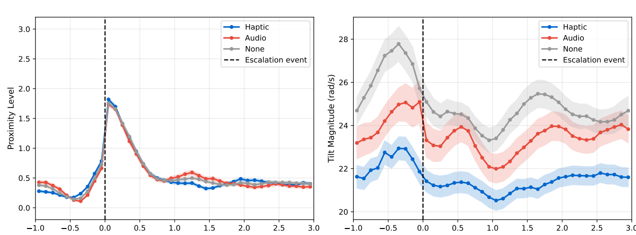

Figure 3: Time-locked proximity level (left) and tilt magnitude (right) relative to proximity escalation events. Vertical dashed line marks the escalation moment ($t = 0$). Events within 500 ms of a fall or respawn were excluded. Haptic (blue), Audio (medium gray), None (dark gray). Shaded bands: ±SE.

Qualitative feedback. Participant P008 (audio) noted: *"The sounds made me nervous. It's a good training for elders or students with special needs."* This is consistent with audio cues increasing arousal without improving control.

5 Discussion

Interpreting the null finding. PathSense detected no significant fall-rate differences across modalities, and Bayesian analysis confirms this is not merely low power: $BF_{01} \approx 84$ at Easy level provides strong evidence that no modality difference exists under these conditions. Effect sizes were small to negligible (rank-biserial r from -0.44 to $+0.25$). This suggests that modality benefits are not automatic but depend on task structure, training duration, or individual factors. In itself, this advances HCI understanding of proximity feedback design [7].

The baseline confound as methodological insight. The 10-fold practice-round gap (audio 0.33 vs. haptic 4.88, $p = 0.04$, $r = -0.57$) points to pre-existing group differences in tilt control ability and motivates baseline-controlled study designs. Change-from-baseline deltas show moderate effect sizes (Cohen's $d \approx 0.5$–0.7) suggesting modality-specific learning trajectories that raw-falls analysis obscures: haptic and none groups improved at Easy while audio did not, consistent with audio cues increasing arousal without aiding correction. At Hard level all groups deteriorated equally ($\Delta = +8$ to $+11$ falls), indicating that task difficulty overrides any modality benefit beyond a certain complexity threshold.

6 Limitations and Future Work

With 8–9 participants per cell, the study is underpowered; a follow-up with $n \approx 50$ per modality would achieve 80% power for medium effects. Most participants were 13–15 years old, while the theoretical background draws on adult motor control and body schema literature; replication with adult samples is essential to separate age effects from modality effects. Random assignment did not equate practice-round performance, so future studies should counterbalance or control the practice phase to avoid the baseline confound observed here. Finally, each participant completed only one session, so learning consolidation and retention were not assessed; multi-session designs are needed to determine whether modality differences emerge with extended practice.

7 Conclusion

PathSense demonstrates a smartphone-native platform for comparing proximity feedback modalities in a tilt-controlled motor task. In this pilot study we found no raw differences in fall rate across modalities, but uncovered a critical baseline confound and evidence that change-from-baseline analysis reveals modality-specific patterns (haptic and none improve at easier levels; audio remains flat). These findings suggest that feedback modality effects may be subtle, task-dependent, and developmentally sensitive. Rather than declaring haptic a clear winner, this work highlights the importance of (i) controlled baselines, (ii) developmental-stage matching in theoretical predictions, and (iii) larger sample sizes for confirmatory claims. The PathSense platform and pilot data provide effect size estimates and design specifications for a larger follow-up.

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